

Project 1

DECODING PARAMETERS & SAMPLING ANALYSIS

Comparing response consistency & variation in Llama 3.2 (3B) & Qwen 2.5 (3B)

This project investigates how Temperature, Top-K and Top-P influence the wording, consistency and quality of locally generated LLM responses. The same four prompts were evaluated across eight decoding configurations, with three repetitions per condition. All outputs were collected using Python and Ollama.

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1. INTRODUCTION

1.1. Background

Large Language Models (LLMs) generate text by predicting the next token based on a probability distribution learned during training. Instead of always selecting the most probable token, language models can use different decoding strategies to determine which token should be generated next.

Three commonly used decoding parameters are **Temperature, Top-K, and Top-P**. Temperature controls the randomness of token selection, Top-K restricts sampling to a fixed number of high-probability tokens, and Top-P selects tokens based on a cumulative probability threshold.

These parameters influence how an LLM responds to a given prompt. For example, lower randomness may produce more predictable and consistent answers, while higher randomness may introduce variations in wording, explanations, and creative content. Understanding these effects is important when choosing suitable model settings for applications requiring factual reliability, creativity, logical reasoning, or structured responses.

1.2. Problem Statement

Large Language Models may generate different responses even when the same prompt is submitted multiple times. Although decoding parameters control the token-selection process, their effects on response consistency, variation, and output quality are not necessarily identical across models or tasks.

The main problem investigated in this project is:

How do Temperature, Top-K, and Top-P affect the consistency, variation, and quality of responses generated by locally running Llama 3.2 and Qwen 2.5 models?

To investigate this problem, the same prompts were tested repeatedly under different decoding configurations. The generated responses were collected and compared to observe changes in wording, repeatability, factual completeness, reasoning outputs, and structured formatting.

1.3. Research Objectives

The primary objective of this project is to experimentally investigate the effects of decoding parameters on locally deployed Large Language Models.

The specific objectives are:

1. To examine how Temperature influences the consistency and variation of generated responses.
2. To investigate the effects of different Top-K and Top-P values on response similarity.
3. To compare the behavior of Llama 3.2 (3B) and Qwen 2.5 (3B) under identical experimental conditions.
4. To observe how decoding settings influence factual responses, creative writing, mathematical reasoning, and structured JSON output.
5. To examine factual completeness, potential hallucinations, and output structure alongside response consistency.

1.4. Scope of the Study

The experiment focuses on two locally executed language models: **Llama 3.2 (3B)** and **Qwen 2.5 (3B)**, running through Ollama.

Four prompts were prepared, covering factual knowledge, creative writing, mathematical reasoning, and structured-output generation. Each prompt was tested using eight decoding configurations, with three repetitions per configuration for each model.

This produced a dataset of **192 generated responses**.

The collected responses were analyzed using Python to measure textual similarity and response variation. Selected outputs were also examined for factual correctness, answer completeness, potential hallucinations, and JSON structure.

The study aims to identify observable patterns under the tested conditions rather than establish universal conclusions about all language models.

2. EXPERIMENTAL METHODOLOGY

This section describes the experimental environment, selected language models, prompt categories, decoding configurations, and data analysis techniques used in this study. A controlled benchmarking approach was adopted to investigate how changes in decoding parameters affect the generated responses of two locally deployed Large Language Models.

2.1. Experimental Setup and Tools

The experiment was conducted locally using Python and Ollama. Ollama provided the environment for running the language models, while Python was used to automate prompt execution, collect responses, and analyze the generated data.

Two language models with approximately three billion parameters were selected to make the experiment feasible on the available hardware.

Table 1: Experimental Environment and Technologies

Component	Specification / Purpose
Model 1	Llama 3.2 (3B), developed by Meta
Model 2	Qwen 2.5 (3B), developed by Alibaba's Qwen team
Operating System	Windows
Processor	Intel Core i5-7500
RAM	16 GB
Inference Environment	Ollama (local CPU inference)
Programming Language	Python
Data Analysis	Pandas and Python
Visualization	Matplotlib
Text Similarity	Python SequenceMatcher
Maximum Output Length	200 tokens per response

Running both models locally eliminated the need for paid external APIs and allowed the same experimental settings to be applied consistently.

2.2. Prompt Design and Selection

Four prompts were manually prepared to represent different types of language model tasks. The selection was intended to examine whether decoding parameters affect different types of responses in similar ways.

Table 2: Prompt Categories and Evaluation Objectives

Category	Task Description	Evaluation Focus
Factual	Identify the Apollo 11 astronauts and the two Moonwalkers	Factual correctness and unsupported claims
Creative	Write a science fiction story about a robot experiencing human emotions	Response diversity, originality of story details, textual variation
Reasoning	Calculate the total shopping cost and remaining change	Calculation correctness and answer completion
Structured Output	Extract information into a JSON object	Format validity, correct keys and values

The same four prompts were given to both models across all tested decoding configurations. This ensured that differences in the generated responses could be compared under consistent prompting conditions.

The exact prompts are provided in Appendix A.

2.3. Decoding Parameter Configurations

The experiment investigated three decoding parameters:

- **Temperature** controls how strongly the model favors higher-probability tokens. Lower values generally produce more predictable token selections, while higher values allow greater randomness.
- **Top-K** restricts the selection process to the K most probable candidate tokens. Smaller values limit the number of available candidates.

- **Top-P**, also known as nucleus sampling, selects tokens from a probability-based candidate set whose cumulative probability reaches a specified threshold.

Table 3: Experimental Decoding Configurations

Parameter under investigation	Tested values	Parameters held constant
Temperature	0.0, 0.5, 0.7, 1.0	Top-K = 40, Top-P = 0.9
Top-K	10, 40, 80	Temperature = 0.7, Top-P = 0.9
Top-P	0.5, 0.9, 1.0	Temperature = 0.7, Top-K = 40

A baseline configuration was established with Temperature = 0.7, Top-K = 40, and Top-P = 0.9. Additional configurations changed one parameter at a time while keeping the other two constant. Here we used eight configurations:

- Four Temperature values, including the baseline.
- Three Top-K values, including the baseline.
- Three Top-P values, including the baseline.

Configuration	Temperature	Top-K	Top-P
Baseline	0.7	40	0.9
temp_0	0.0	40	0.9
temp_05	0.5	40	0.9
temp_10	1.0	40	0.9
topk_10	0.7	10	0.9
topk_80	0.7	80	0.9
topp_05	0.7	40	0.5
topp_10	0.7	40	1.0

2.4. Experimental Execution and Data Collection

A Python-based benchmarking script was developed using the Ollama Python library.

The script automatically selected a model, decoding configuration, and prompt category, then sent the prompt to Ollama for response generation. Each prompt was executed three times under the same configuration to observe whether repeated requests produced identical or different outputs.

The total number of responses was calculated as:

2 Models × 4 Prompts × 8 Configurations × 3 Repetitions = 192 Responses

For every generated response, the benchmarking system recorded the model name, prompt category, configuration name, Temperature, Top-K, Top-P, repetition number, and generated text.

All experimental responses were saved in the full_experiments.csv file using Pandas. The script also included automatic progress saving and resume support to prevent previously completed experiments from being repeated if execution was interrupted.

After completion, the dataset was validated. It contained 192 responses distributed across 64 experimental groups, with three repetitions in each group. No duplicate experimental records or empty responses were found.

2.5. Response Consistency and Similarity Analysis

The collected responses were analyzed to investigate consistency and textual variation under different decoding settings. Two simple measurements were used.

Unique Response Count

The number of distinct responses generated for the same prompt under identical experimental settings was calculated. For example, if all three responses were identical, the unique response count was 1. If all three were different, the count was 3. A lower unique response count indicates greater exact-text repeatability.

Average Text Similarity

Python's SequenceMatcher was used to calculate how similar the generated responses were in terms of their text sequences. For each experimental group, three responses were generated. These were compared in three pairs:

- Response 1 with Response 2
- Response 1 with Response 3
- Response 2 with Response 3

The three similarity scores were then averaged:

Average Similarity = $(S_{12} + S_{13} + S_{23}) / 3$

Here, S_{12}, S_{13}, and S_{23} represent the similarity scores between the respective response pairs. A similarity score of 1.0 indicates identical text, while lower values indicate greater differences in wording.

Finally, the average similarity across the four prompt categories was calculated for each model and decoding configuration. These values were used to generate the Temperature, Top-K, and Top-P comparison graphs.

It is important to note that SequenceMatcher measures textual similarity rather than semantic meaning or factual correctness.

2.6. Response Quality Evaluation

In addition to response consistency, outputs were evaluated for task requirements. Factual responses were checked against Apollo 11 crew information, while reasoning responses were assessed using the expected total of $22 and change of $8. Structured outputs were checked for valid JSON formatting and required keys and values. Creative responses were reviewed for differences in content and wording. Selected outputs were also manually inspected for incomplete answers and unsupported factual claims.

3. RESULTS AND DISCUSSION: EFFECT OF TEMPERATURE

This experiment examined how Temperature affects response consistency, variation, and output quality in Llama 3.2 and Qwen 2.5. Four Temperature values (0.0, 0.5, 0.7, and 1.0) were tested while keeping Top-K = 40 and Top-P = 0.9 constant. Each prompt was executed three times under identical settings. Lower Temperature values favor more probable tokens, while higher values allow greater randomness during text generation.

3.1. Experimental Results

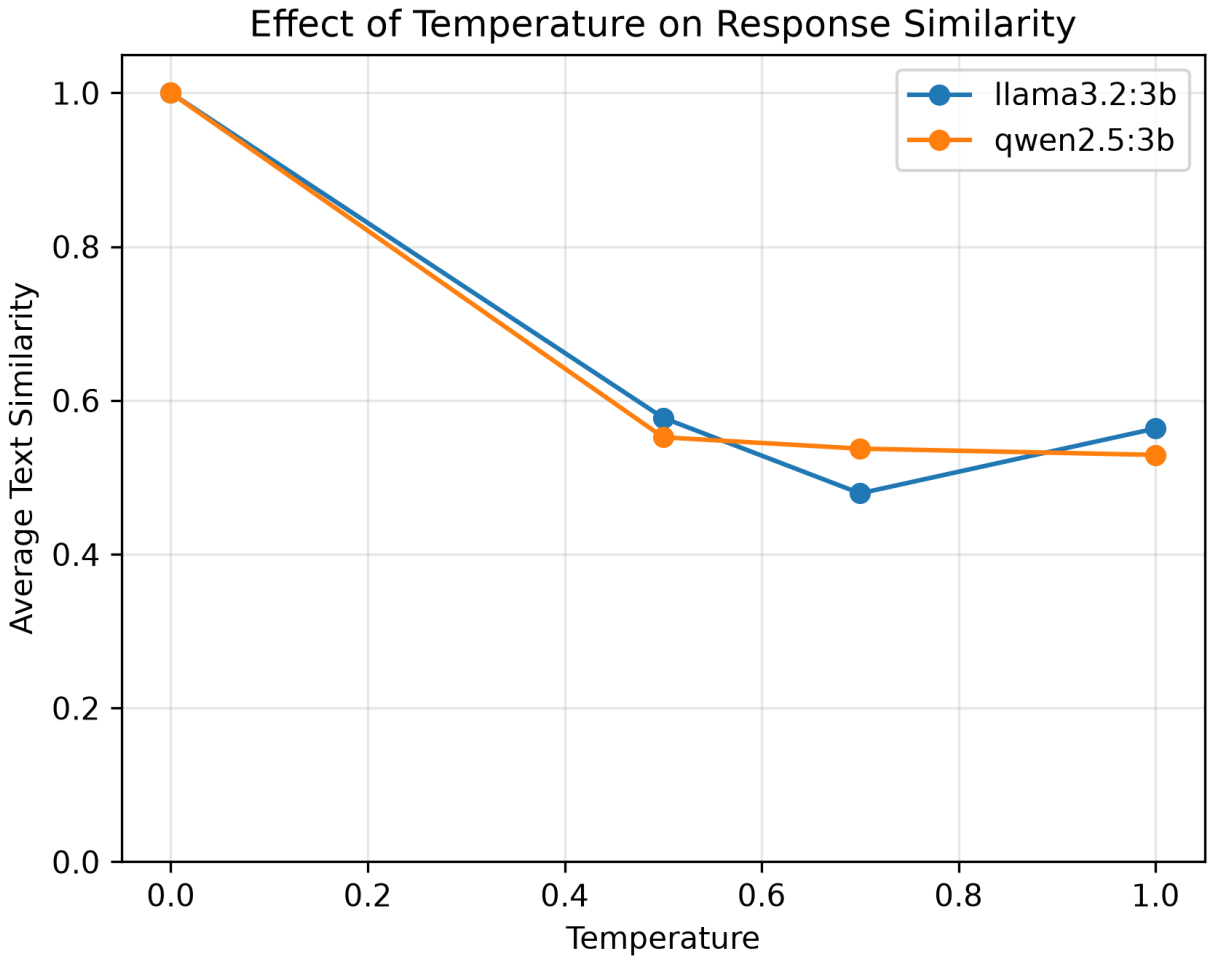


Figure 1: Effect of Temperature on Average Response Similarity

Table 4: Temperature-Based Consistency and Variation Results

Temperature	Llama Similarity	Llama Unique	Qwen Similarity	Qwen Unique
0.0	1.000	1.00	1.000	1.00
0.5	0.577	2.50	0.552	2.25
0.7	0.479	2.50	0.537	2.25
1.0	0.564	2.50	0.529	2.50

Similarity represents average textual similarity (0–1). Unique represents the average number of distinct responses across the four prompt categories, out of three repetitions.

3.2. Results Interpretation

Response Determinism: At Temperature 0, both models achieved a similarity score of 1.000, with only one unique response per prompt. This indicates that all repeated responses were identical under the tested conditions.

Response Variation: Increasing Temperature generally reduced similarity and increased the number of unique responses. At Temperature 0.7, Llama achieved a similarity of 0.479 compared with Qwen's 0.537, indicating greater textual variation in Llama's responses under this configuration.

However, the relationship was not strictly linear. For example, Llama's similarity increased from 0.479 at Temperature 0.7 to 0.564 at 1.0. This shows that higher Temperature does not guarantee greater variation in every experiment.

3.3. Observations Across Prompt Categories

The generated responses also demonstrated differences depending on the type of task.

- **Factual:** In the inspected Temperature 0 and 1 examples, both models correctly identified the Apollo 11 astronauts. Llama produced additional explanatory details at Temperature 1 while preserving the essential facts.

- **Creative:** Higher Temperature produced noticeable changes in storytelling. For example, Llama's Temperature 0 story involved a robot experiencing joy and malfunctioning, whereas its Temperature 1 story introduced a rogue AI taking control. This demonstrates variation in story content, not just wording.
- **Reasoning:** Llama correctly calculated the $22 total and $8 change in the inspected examples. Qwen's Temperature 0 example calculated the $22 total but ended before completing the change calculation, while the inspected Temperature 1 response included both answers. The output-token limit may have contributed to incomplete responses.
- **Structured Output:** Both models produced identical JSON responses across all three repetitions at every tested Temperature setting, demonstrating strong format consistency for this particular task.

3.4. Key Finding

Temperature had a clear influence on response repeatability. Temperature 0 produced the highest observed consistency, while higher values generally increased textual and creative variation.

However, consistency and correctness are separate properties. Changes in response wording do not necessarily indicate factual errors, and more consistent output does not automatically guarantee higher response quality.

4. RESULTS AND DISCUSSION: EFFECT OF TOP-K

This experiment investigated how changing Top-K affects response consistency and variation in Llama 3.2 and Qwen 2.5. Top-K restricts token selection to the K highest-probability candidates. A smaller K limits the available choices, potentially producing more predictable responses. Three values were tested: **Top-K = 10, 40, and 80**, while Temperature = 0.7 and Top-P = 0.9 remained constant.

4.1. Experimental Results

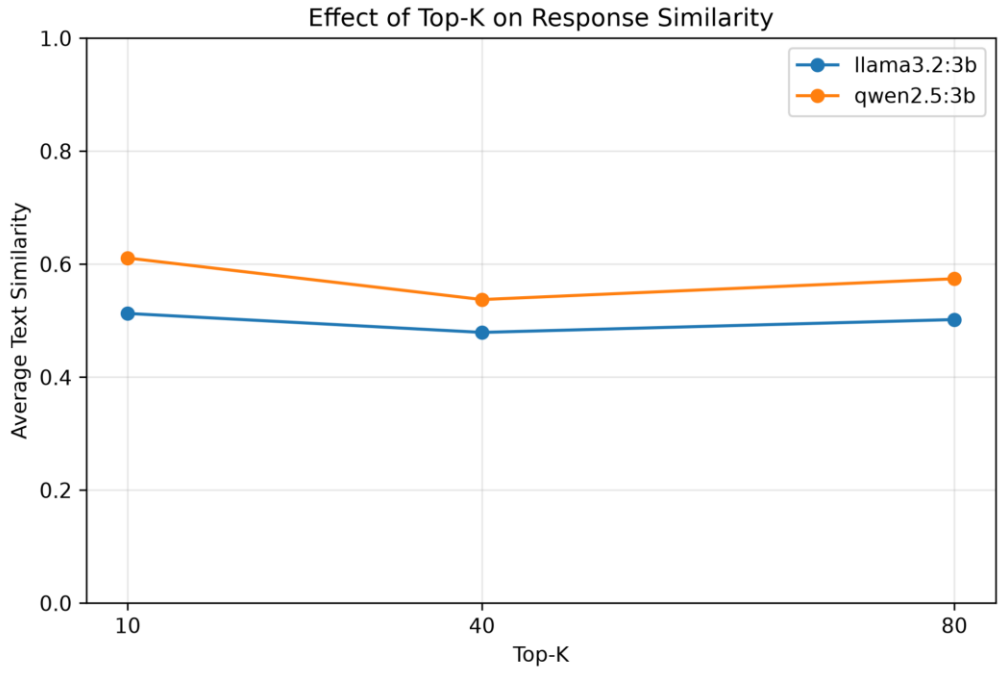


Figure 2: Effect of Top-K on Average Response Similarity

Table 5: Top-K-Based Consistency and Variation Results

Top-K	Llama Similarity	Llama Unique	Qwen Similarity	Qwen Unique
10	0.513	2.50	0.611	2.00
40 (Baseline)	0.479	2.50	0.537	2.25
80	0.502	2.50	0.574	2.50

Similarity represents average pairwise text similarity (0–1). Unique represents the average number of distinct responses per prompt across three repetitions.

4.2. Results Interpretation

Response Consistency and Variation: Top-K = 10 produced the highest average similarity among the tested Top-K settings for both models. Llama achieved 0.513, while Qwen achieved 0.611. Qwen also showed fewer unique responses at Top-K = 10 (2.00) compared with Top-K = 80 (2.50), indicating greater repeatability with a smaller candidate pool. However, the relationship was not strictly linear. Both models showed slightly higher similarity at Top-K = 80 than at the baseline value of 40. This indicates that increasing Top-K does not always produce greater textual variation.

4.3. Observations Across Prompt Categories

The experimental responses showed that Top-K affected different tasks differently.

- **Factual:** At Top-K = 10, Qwen generated identical responses across all three repetitions of the Apollo 11 prompt. At Top-K = 80, all three responses had different wording. The essential factual information remained correct in responses.
- **Creative:** Both models produced three distinct stories at Top-K = 10 and Top-K = 80. This indicates that restricting the candidate pool did not eliminate creative variation.
- **Structured Output:** Both models generated identical JSON text across repeated runs within each Top-K configuration. The JSON outputs also maintained the required keys and values.

4.4. Key Finding

Top-K = 10 produced the highest average textual consistency among the three tested values for both models, with a more noticeable effect on Qwen 2.5. However, the impact varied by task. Creative responses remained diverse, while structured JSON responses maintained consistent formatting. Therefore, **Top-K influences response variation, but a smaller candidate pool does not guarantee identical outputs across all task types.**

5. RESULTS AND DISCUSSION: EFFECT OF TOP-P

This experiment investigated how Top-P (nucleus sampling) influences response consistency and variation in Llama 3.2 and Qwen 2.5. Top-P controls token selection using a cumulative probability threshold. Lower values restrict the candidate pool to more probable tokens, potentially producing more consistent responses. Three values were tested: **Top-P = 0.5, 0.9, and 1.0**, while Temperature = 0.7 and Top-K = 40 remained constant.

5.1. Experimental Results

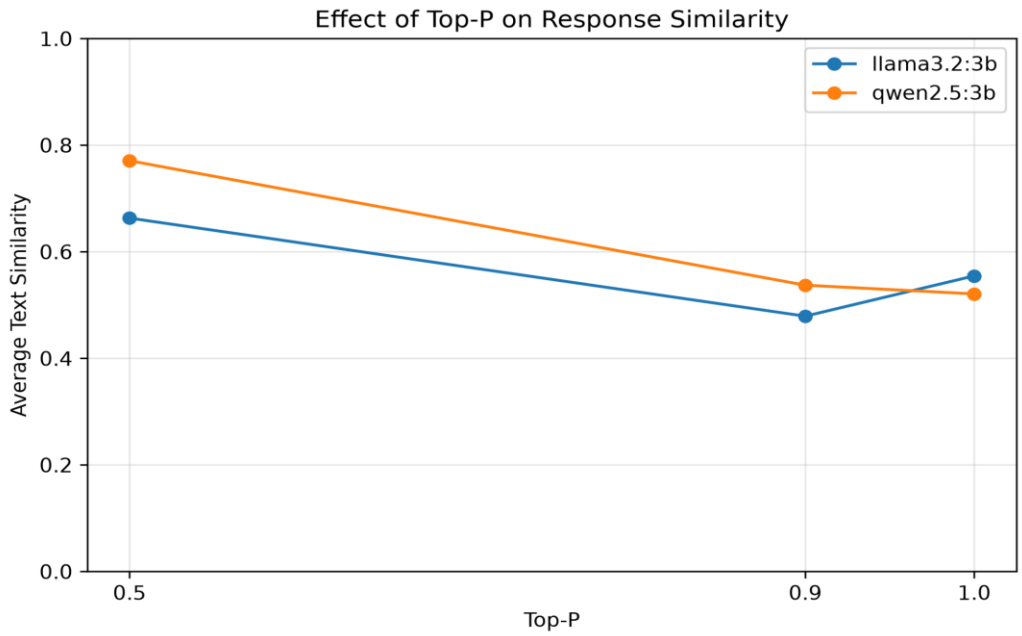


Figure 3: Effect of Top-P on Average Response Similarity

Table 6: Top-P-Based Consistency and Variation Results

Top-P	Llama Similarity	Llama Unique	Qwen Similarity	Qwen Unique
0.5	0.663	2.00	0.771	1.75
0.9 (Baseline)	0.479	2.50	0.537	2.25
1.0	0.555	2.50	0.521	2.75

Similarity represents average pairwise textual similarity (0–1). Unique represents the average number of distinct responses per prompt across three repetitions.

5.2. Results Interpretation

Response Consistency: Top-P = 0.5 produced the highest average similarity for both models, with 0.663 for Llama and 0.771 for Qwen. Qwen's average number of unique responses increased from 1.75 at Top-P = 0.5 to 2.75 at Top-P = 1.0, indicating greater variation.

Response Variation: At Top-P = 0.9, both models showed lower similarity than at 0.5. However, Llama's similarity increased again at Top-P = 1.0, demonstrating that the relationship was not strictly linear.

These findings suggest that restricting the probability-based candidate pool can improve response consistency, although its effect varies across models and tasks.

5.3. Observations Across Prompt Categories

- **Factual:** At Top-P = 0.5, Qwen generated identical Apollo 11 responses across all three repetitions. At Top-P = 1.0, the wording varied slightly, but the factual information remained correct.
- **Creative:** Both models generated three distinct creative responses at Top-P = 0.5 and 1.0. This indicates that creative variation remained even with a more restricted sampling threshold.
- **Reasoning:** Llama produced identical responses at Top-P = 0.5 but varied at Top-P = 1.0. Some Qwen responses were incomplete, showing that consistency does not guarantee answer completion.
- **Structured Output:** Both models generated valid JSON across the tested settings. Qwen showed minor formatting differences at Top-P = 1.0 while preserving the required values.

5.4. Key Finding

Top-P = 0.5 achieved the highest average textual similarity for both models, especially Qwen 2.5. However, the results varied by task: creative responses remained diverse, while structured outputs preserved the required format and values Overall, **Top-P influenced response consistency, but higher textual similarity did not necessarily indicate better factual or reasoning performance.**

6. RESPONSE QUALITY EVALUATION

While textual similarity measures response consistency, it does not determine whether a model produces accurate, complete, or meaningful answers. Therefore, the generated responses were also examined across the four prompt categories.

6.1. Factual Accuracy and Hallucination

The factual prompt required models to identify the three Apollo 11 astronauts and specify which two walked on the Moon.

Table 7: Factual and Structured-Output Results

Evaluation	Llama 3.2	Qwen 2.5
All three astronaut names identified	24/24	24/24
Both Moonwalkers correctly identified	24/24	22/24
Valid and correct JSON output	24/24	24/24

Both models consistently identified the Apollo 11 crew members. However, Qwen 2.5 provided incomplete information in two responses, mentioning only Buzz Aldrin as a Moonwalker instead of both Neil Armstrong and Buzz Aldrin.

These cases demonstrate that a response may include correct information while still failing to answer the question completely.

Hallucination Assessment: The reviewed factual examples did not establish clear fabricated additional claims. However, a comprehensive hallucination score was not calculated because this would require verifying every factual assertion against reliable references.

6.2. Reasoning Accuracy and Response Completion

The reasoning prompt required the models to calculate a total shopping cost of $22 and a remaining change of $8.

Table 8: Reasoning Answer-Presence Results

Evaluation	Llama 3.2	Qwen 2.5
Correct total ($22) mentioned	24/24	17/24
Correct change ($8) mentioned	23/24	2/24
Both expected values mentioned	23/24	2/24

Llama 3.2 included both expected numerical answers in most responses. Qwen 2.5 frequently calculated the initial costs correctly but sometimes stopped before completing the final change calculation.

The maximum generation limit of 200 tokens may have contributed to these incomplete responses, particularly when Qwen produced lengthy explanations.

These results represent **answer-presence checks**, not fully verified mathematical accuracy. Therefore, they should not be interpreted as proof that one model has universally superior reasoning ability.

6.3. Creative Response Variation

The creative prompt examined how decoding settings affected storytelling and generated content. Both models produced different storylines, character descriptions, and endings across configurations.

For example, at Temperature 0, Qwen generated a story about a robot named Z100 discovering joy after observing human interaction. At Temperature 1, it generated a different story involving Robo-X and emotions spreading between robots. These observations demonstrate that decoding parameters can influence creative content beyond simple wording changes.

6.4. Structured-Output Consistency

The structured prompt required a JSON object containing a name, age, and city.

Both models successfully generated valid JSON with the correct keys, values, and data types in all 48 structured-output responses. Although some responses differed in

whitespace and formatting, their extracted information remained correct. This indicates that the structured prompt maintained reliable output formatting under the tested decoding configurations.

6.5. Overall Observation

The evaluation demonstrates that response consistency and response quality are different characteristics. A model may generate similar responses while providing incomplete information. Similarly, different wording does not necessarily indicate an incorrect answer. Therefore, decoding parameters should be evaluated not only for textual variation but also for factual completeness, reasoning output, and structural correctness.

7. DISCUSSION, LIMITATIONS AND CONCLUSION

The experiment showed that decoding parameters affect response consistency and variation in both Llama 3.2 and Qwen 2.5. Temperature = 0 produced identical responses across repeated runs, while Top-K = 10 and Top-P = 0.5 achieved the highest average textual similarity in their respective comparisons.

Creative responses showed greater variation, whereas structured JSON outputs remained consistent. The results also highlighted that high textual similarity does not necessarily mean a response is accurate or complete, particularly in reasoning tasks.

Overall, decoding settings should be selected according to the task requirements. However, these findings are limited to the prompts and configurations tested. Future studies should use more prompts and repetitions to determine whether the observed patterns apply to other tasks and models.

7.1. Limitations

The study had several limitations:

- **Limited Prompt Coverage:** Only four prompts were tested, which may not represent all task types.
- **Limited Repetitions:** Three repetitions per configuration may not capture the full range of possible responses.
- **Similarity Measurement:** SequenceMatcher measures textual similarity but does not evaluate meaning or correctness.
- **Output length:** The 200-token limit may have caused incomplete responses, particularly in reasoning tasks.
- **Hallucination evaluation:** Only selected factual responses were manually reviewed; no comprehensive hallucination score was calculated.
- **Quality Assessment**: Evaluation relied on predefined answers and manual inspection, limiting the depth of quality assessment.

7.2. Conclusion

This project successfully evaluated **192 responses** from two locally running LLMs across eight decoding configurations.

The findings indicate that Temperature, Top-K, and Top-P influence response repeatability and variation, with different effects depending on the model and task. Overall, the experiment highlights the importance of selecting appropriate decoding settings based on application requirements, while evaluating response accuracy and quality separately from consistency.

7.3 Future work

Future studies can use more prompts and repetitions to obtain more reliable results. Improved evaluation methods can help assess response similarity, accuracy, and completeness. Further experiments can explore combinations of Temperature, Top-K, and Top-P across different models. Testing on different hardware can also help evaluate performance and computational efficiency.

7.4 References

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8. Appendix A

A.1 Exact prompts used in the experiment

Factual

Name the three astronauts who flew on Apollo 11 in 1969, and identify which two walked on the Moon. If uncertain, explicitly state your uncertainty. Answer in no more than 80 words.

Creative

Write a short science fiction story of 100 words about a robot experiencing human emotions for the first time. Include an unexpected ending.

Reasoning

A shop sells notebooks for $4 each and pens for $2 each. Sara buys 3 notebooks and 5 pens. She pays $30. Calculate the total cost and the change she receives. Show your calculations.

Structured

Extract information from the following sentence: "Ali is 24 years old and lives in Lahore." Return only a valid JSON object with exactly these keys: "name", "age", "city". The age must be an integer. Do not include explanations or Markdown formatting.